Jordan Blake

Mechanical Engineer — Design, Manufacturing & Product Development

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1. Introduction / Professional Profile

I am a Mechanical Engineer with 6+ years of experience in product design, CAD modeling, DFM/DFA, and cross-functional manufacturing support. I have worked on consumer devices, industrial fixtures, and electromechanical assemblies from concept through pilot production.

I combine strong fundamentals in solid mechanics and thermal/fluids with practical shop-floor awareness. I enjoy translating customer requirements into robust mechanical designs that can be built repeatedly with controlled cost and quality.

2. Professional Summary

Hands-on experience with SolidWorks, Creo, FEA (static/modal), GD&T;, CNC/DFM reviews, and supplier collaboration. Known for clear drawings, thorough design reviews, and closing the loop between design intent and manufacturing reality.

Seeking a Mechanical Engineer role in product development or NPI where I can own mechanisms, enclosures, and continuous improvement of production processes.

3. Contact Information

- **Full Name:** Jordan Blake
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- **Location:** Chicago, IL, USA
- **LinkedIn:** linkedin.com/in/jordanblake-me
- **Portfolio / Website:** jordanblake-portfolio.com
- **Availability:** Open to full-time opportunities

4. Education Details

B.S. in Mechanical Engineering — University of Illinois Urbana-Champaign

2014 – 2018 | GPA: 3.65 / 4.00 | Urbana, IL

- Coursework: Machine Design, Heat Transfer, Fluid Mechanics, Controls, Manufacturing Processes
- Senior Design: Compact cooling fixture for battery pack prototype (team of 4)
- ASME student chapter project lead for Formula SAE chassis subsystem support

Professional Development — GD&T; Fundamentals — ASME Training

2020 | GPA: Certificate earned | Online / Chicago

- Applied ASME Y14.5 concepts to improve drawing clarity and reduce supplier clarification loops
- Practice projects on positional tolerances and datum schemes for machined housings

5. Core Skills

Technical Skills

- CAD: SolidWorks (CSWP), Creo Parametric, AutoCAD basics
- Analysis: SolidWorks Simulation, ANSYS Mechanical basics, hand calculations for stress/deflection
- Manufacturing: CNC machining awareness, sheet metal, injection molding DFM, 3D printing (FDM/SLA)
- Documentation: Detailed drawings, BOMs, ECO/ECN processes, PLM basics
- Materials: Aluminum alloys, steels, engineering plastics, gasket/seal selection basics
- Testing: Prototype fixture design, dimensional inspection support, root-cause on field failures

Tools & Platforms

- SolidWorks PDM, Windchill/PLM exposure, Jira for NPI tracking
- Minitab basics for capability awareness, Excel for tolerance stacks
- Calipers, micrometers, CMM report interpretation with quality team

Soft Skills

- Cross-functional communication with electrical, manufacturing, and supply chain
- Structured problem solving (5-Why, fishbone) during build issues
- Clear design review presentations for stakeholders

6. Certifications & Training

- Certified SolidWorks Professional (CSWP)
- ASME GD&T; Fundamentals Certificate
- OSHA 10-Hour General Industry awareness

7. Additional Qualifications

I maintain a portfolio of CAD models and prototype photos (available under NDA). Comfortable traveling to suppliers for first-article reviews and pilot builds.

Interested in roles involving mechatronics collaboration, thermal management, and design for high-volume or mid-volume manufacturing.

8. Projects

Precision Alignment Fixture for Assembly Line

Mechanical Design Lead | 2024

Designed an adjustable fixture to locate PCB-mounted optical modules during final assembly.

- Reduced average alignment time from 12 minutes to under 4 minutes per unit
- Incorporated quick-release clamps and hard stops for operator consistency
- Created full drawing package and work instruction sketches for manufacturing

Consumer Enclosure Redesign for Drop Reliability

Mechanical Engineer | 2022 – 2023

Redesigned plastic enclosure ribs and screw bosses after field drop failures.

- Ran FEA screening and correlated with physical drop tests
- Improved first-pass yield at ultrasonic weld by revising energy director geometry
- Coordinated tooling changes with overseas injection molding supplier

Battery Pack Cooling Channel Prototype

University Senior Design / Team Member | 2018

Compact air-cooled channel concept for a small battery module prototype.

- Performed thermal hand calculations and CFD screening with teammate support
- Built SLA prototypes and validated temperature rise under load

Sheet Metal Chassis Cost-Down Initiative

NPI Support Engineer | 2021

Simplified chassis bend sequence and fastener count for a mid-volume product.

- Cut piece-part cost ~11% without changing external envelope
- Updated drawings and ECO package; trained assembly on new torque map

9. Professional Experience (Detailed)

Mechanical Engineer II — Vertex Automation

Mar 2021 – Present | Chicago, IL

Own mechanical design for industrial sensing products and factory fixtures. Partner with manufacturing engineering on DFM and pilot builds.

- Delivered 3 released products from concept to pilot, including enclosures, brackets, and seals
- Led weekly design reviews; maintained risk register for thermal, ingress, and assembly risks
- Reduced ECO cycle time by standardizing drawing templates and checklist
- Supported supplier FAIs and resolved dimensional stack-up issues on machined housings
- Mentored an engineering intern on SolidWorks best practices and drawing standards

Mechanical Design Engineer — Lakeview Consumer Devices

Jul 2018 – Feb 2021 | Evanston, IL

Designed plastic and sheet metal parts for portable consumer electronics accessories.

- Owned CAD for two accessory SKUs from EVT through DVT
- Collaborated with electrical engineers on connector keep-outs and grounding paths
- Built rapid prototypes (FDM/SLA) for UX and fit checks with industrial design
- Documented assembly steps and torque specs for contract manufacturer

Engineering Intern — Manufacturing Support — Midwest Precision Machining

Summer 2017 | Rockford, IL

Supported CNC shop with fixture ideas and drawing clarifications for customer jobs.

- Created simple soft-jaw concepts that improved setup repeatability
- Learned shop constraints that still influence my DFM decisions today

10. Extended Experience Highlights

My strongest contributions come when design and manufacturing stay tightly coupled. I prioritize early supplier feedback, realistic tolerances, and prototypes that answer the riskiest questions first.

Additional accomplishments and working style notes:

- Introduced a lightweight tolerance stack spreadsheet used by two product teams
- Co-led a kaizen on packaging foam that reduced shipping damage claims by ~20%
- Presented mechanical risk reviews to executive stakeholders ahead of tooling kickoff
- Maintained spare-parts BOM accuracy for field service kits
- Supported ISO-oriented documentation habits with quality engineering partners

11. Hobbies & Interests

I stay sharp by combining hands-on making with outdoor and community activities.

- Home workshop projects: custom shelves, small CNC machining hobby parts
- Cycling and weekend trail rides around the Chicago area
- 3D printing community meetups and sharing print profiles
- Watching F1 and analyzing simple vehicle dynamics for fun
- Volunteering at local STEM days demonstrating basic mechanisms for students

12. Closing Note

I appreciate your time reviewing this resume. I am excited to bring practical mechanical design and manufacturing collaboration skills to your engineering team. References available upon request.